FOLLOW-UP NOTE TO `BT_BASELINE_NOTE` V1.0: $\deg a = 0$ ROW EXTENSION AND V_{quad} ROW-MEMBERSHIP RE-ATTRIBUTION UNDER THE EXTENDED FOUR-ROW ENUMERATION

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ABSTRACT. This note is an additive follow-up to `bt_baseline_note` v1.0 (Zenodo concept DOI 10.5281/zenodo.20048196; version DOI 10.5281/zenodo.20048197). The v1.0 main result ([5, Theorem 1.1]) is canonical as stated for the band $\deg a \in \{d-1, d, d+1\}$ of three SIARC conventions ([5, §2.2 – §2.3]). The present note records two additive items, both grounded entirely in pre-existing substrate: (i) the Phase A WZ-balance table extends to a fourth row at $\deg a = 0$ (the $(1, b)$ corner inhabited by every PCF-2 harvest implementation per LANE-2 P1), with $A_{\mathrm{naive}} = 2d$ at that corner via Balance III; and (ii) the V_{quad} entry of the PCF-1 v1.3 §6 Theorem 5 row table aligns with the $\deg a = 0$ row at $d = 2$ as a matter of row-membership, so the open content of v1.0 §6 Q [mechanism] (the closure of the borderline-locus posit (i′)) admits a row-membership reading. The substrate is the LANE-2 canonical re-arbitration (siarc-relay-bridge [11]) together with three implementation reports: 064 [10], 065 [8], and 066 [7]. The v1.0 `.tex` source is unmodified by this note.

1. Scope and relationship to v1.0

This follow-up note is the canonical implementation of the LANE-2 Item 3 verdict `LEAVE_V1_0_CANONICAL_WITH_V` ([4, Item 3 reasoning, L173–200]). Per that verdict, `bt_baseline_note` v1.0 Theorem 1.1 is canonical as stated for the band $\deg a \in \{d-1, d, d+1\}$, and the deg_a = 0 row described below is an *additive extension*, not a correction. The v1.0 `.tex` source (file SHA-256 `6746692C517DC25238473E819527C5682465CDC` 38,023 bytes; deposited at `siarc-relay-bridge/sessions/2026-05-06/T1-PHASE2-BASELINE-NOTE/bt_baselin` is unchanged by the present note. The Zenodo v1.0 PDF (file SHA-256 `23022F0DE77AC8388ED584B2196C0AB995CD8` 409,337 bytes; concept DOI 10.5281/zenodo.20048196; version DOI 10.5281/zenodo.20048197) remains the canonical citation target.

The v1.0 §4.2 prose at L481–487 of the source file describes the V_{quad} upper-branch case as *consistent with the borderline-locus mechanism* (i′) *of* ***`prop:gap-framing`****, the closure of which is the open content of §6 Q [mechanism]* ([5, L481–487]). That prose is forward-looking open-conjecture content as v1.0 itself flags it; the present note resolves the open content of §6 Q [mechanism] via a row-membership reading under the extended four-row Phase A enumeration of relay 064 [10], with no edit to the v1.0 source.

2. Phase A WZ-table extension to $\deg a = 0$

The original Phase A WZ-balance enumeration of v1.0 covers the three SIARC conventions α / symmetric / δ at $\deg a \in \{d-1, d, d+1\}$, $d \in [2, 8]$ ([5, §2.3, §3.4]). Relay 064 records the WZ Newton-polygon balance at the corner case $\deg a = 0$ – the $(1, b)$ continued-fraction corner used by every PCF-2 harvest implementation per LANE-2 P1 ([10, §2.2, §4]). The extension is a single new row at each d and is recorded verbatim in 064 §2.3 as the bold rows of the twelve-row enumeration (file SHA-256 `80E28568FF142B1A81C7C443368349A07A8235CE122D6C6140F88AA0D0706097`, 16,792 bytes).

The four-row enumeration $\deg a \in \{0, d-1, d, d+1\}$ at each d takes the form (reproduced verbatim from [10, §2.3, L102–118]):

Date: 2026-05-07.

d	Convention	$\deg a$	$\deg b$	μ_{dom}	μ_{sub}	A_{naive}
2	$(1, b)$ corner	0	2	2	-2	**4**
2	α	1	2	2	-1	3
2	symmetric	2	2	2	0	2
2	δ	3	2	2	1	1
3	$(1, b)$ corner	0	3	3	-3	**6**
3	α	2	3	3	-1	4
3	symmetric	3	3	3	0	3
3	δ	4	3	3	1	2
4	$(1, b)$ corner	0	4	4	-4	**8**
4	α	3	4	4	-1	5
4	symmetric	4	4	4	0	4
4	δ	5	4	4	1	3

Table 1. The extended four-row Phase A WZ-balance enumeration at $d \in \{2, 3, 4\}$, reproduced verbatim from [10, §2.3, L102–118]. Bold rows are the new $\deg a = 0$ rows added by relay 064; the other rows are reproduced verbatim from the original Phase A summary at `phase_a_summary.md` L34–44 via LANE-2 P2 substrate [2, L77–87].

The new bold rows are derived in LANE-2 V6 ([3, §V6, L210–308], file SHA-256 `56063BF7BA8AD6A01A89FA30A3C6` 15,695 bytes) as Balance III applied at the $(1, b)$ corner ($c_a = 1$, $d_a = 0$). The relevant V6 conclusion is quoted verbatim by 064 §3 ([10, §3]):

> "$A_{\text{naive}} = \mu_{\text{dom}} - \mu_{\text{sub}} = d - (-d) = 2d$ (when $\deg a = 0$). ... General formula: $A_{\text{naive}} = 2d - d_a$." ([3, V6 Step 4, L274–282])

For $d = 2$, $\deg a = 0$: $A_{\text{naive}} = 4$. For $d = 3$: $A_{\text{naive}} = 6$. For $d = 4$: $A_{\text{naive}} = 8$. The sign of the recessive root in the same balance is recorded verbatim in V6 Step 3 at L260–266 of the same file as $\gamma_{\text{sub}} \propto -c_a/c_b$, with the corner specialisation $\gamma_{\text{sub}} = -1/c_b \cdot e^d$ at $c_a = 1, d_a = 0$.

3. V_{quad} as $\deg a = 0$ row member at $d = 2$

The V_{quad} representative of the PCF-1 v1.3 §6 Theorem 5 row table ([6]) carries the empirical exponent $A = 4$, with α-prediction match to 13 digits at dps = 2200 ([6, L566–577], reproduced verbatim in [7, §2.2]). The V_{quad} coefficient declaration in the SIARC audit script `algebraic_independence_audit.py` L37–40 reads `VQUAD_ALPHA = [1]` ([3, V5, L161–179]; reproduced in [7, §3]), so $a_n = 1$ for all n at the source-code declaration level, hence $\deg a = 0$.

LANE-2 P3 records the substrate-level identification: the V_{quad} family has $a_n \equiv 1$ at the source-code level ([2, L77–96]). Combined with Table 1, the empirical $A = 4$ entry of V_{quad} at $d = 2$ aligns row-by-row with the $\deg a = 0$ row of the four-row Phase A enumeration. This row-membership re-attribution is the canonical content of relay 066 ([7, §5]; file SHA-256 `79933B694DD2BF99793429B1A122A4BFF3260A42A93680886D5EF89B4E10FDCD`, 24,073 bytes).

The framing throughout is *row-membership re-attribution under extended enumeration*, additive to v1.0. The PCF-1 v1.3 §6 Theorem 5 row-table source remains canonical; an explicit PCF-1 v1.4 amendment is forward-pointed under the `PCF1-V13-RECONCILE` jurisdiction ([7, §1, §6]) and is not addressed by the present note.

4. PCF-2 cf_value implementation uniformity audit

Relay 065 audits 13 cf_value-class evaluation impls across the `pcf-research/pcf2/` layer (a strictly stronger coverage than the LANE-2 P1 enumeration of nine impls [2, L37–74]). The aggregate verdict is `UNIFORM` at the effective-use layer: 12 of 13 impls classify as `HC1` (strict inline `mp.mpf(1)` for a_n); 1 of 13 classifies as `HC0` (`evaluate_cf` parameterised signature with default λ and zero non-default call sites); 0 of 13 classifies as `PARAM` ([8, §4]; file SHA-256 `16512BCC71C9A19ED59B808D8D070877F2` 11,700 bytes). At the implementation layer of every PCF-2 R1.1 + R1.3 + Q1 harvest pipeline, the $\deg a = 0$ hardcoding is uniform; this is consistent with the V_{quad} identification of Section 3 and supports a uniform $\deg a = 0$ row-membership reading at the $d = 3$ and $d = 4$ strata as well, alongside the empirical $A_{\text{fit}} \approx 2d$ records at those strata ([9] R1.1 + R1.3 + Q1; reproduced in [5, §4.1]).

5. Mechanism (i′) open-content closure under the $\deg a = 0$ row

The v1.0 open-content sentence (verbatim). The open content of v1.0 §4.2 reads, at L481–487 of `bt_baseline_note.tex` ([5, L481–487]):

> "The V_{quad} upper branch is consistent with the borderline-locus mechanism (i′) of `prop:gap-framing` – the closure of which is the open content of §6 Q [mechanism]."

The v1.0 source phrases this as conditional, forward-looking open-content language; the closure of mechanism (i′) on the SIARC stratum is flagged in v1.0 §6 as an open problem (`q:mechanism`; [5, §6]).

Closure under the $\deg a = 0$ row reading. Under the extended four-row enumeration of Table 1, the V_{quad} entry $A = 4$ at $d = 2$ aligns with the $\deg a = 0$ row of the Phase A table, with $A_{\text{naive}} = 2d$ via Balance III at the corner $c_a = 1$, $d_a = 0$ (Section 2). This row-corner placement is independent of any borderline-locus posit: the $\deg a = 0$ row is a normal-case Balance III row, and Balance III is one of the three explicit balance branches of the v1.0 Wimp–Zeilberger enumeration ([5, §3.2]). A concrete consequence: under the row-membership reading, the v1.0 §6 open problem `q:mechanism` admits a closure in which V_{quad}'s upper branch is realised as a row-corner of the extended Phase A enumeration, rather than as a borderline-locus phenomenon distinct from row-membership. The borderline-locus posit (i′), considered as a candidate structural mechanism on *other* SIARC strata where the $\deg a = 0$ row reading does not apply, remains an open future inquiry; the present note does not address it.

Naming the gap. The descriptive term "protocol-to-stratum mismatch" (the gap between the protocol-declared $\deg a$-domain of PCF-2 v1.3 §6 – $\deg a \in \{0, 1\}$ – and the actually-evaluated $\deg a$ at the implementation layer, uniformly $\deg a = 0$ per Section 4) is a candidate label for the gap that the row-membership reading addresses;[1] the substrate basis for the term is LANE-2 V3 + V4 + V5 + P4 ([3, §V3 – §V5], [2, §P4]).

6. Limitations and open questions

The row-membership reading of Section 3 and Section 5 carries the following deliberate scope boundaries. None of the items below is closed by the present note.

- The $\deg a = 0$ row reading identifies the formal-level baseline $A_{\text{naive}} = 2d$ at the $\deg a = 0$ corner under Phase A enumeration. The empirical record carried by [9] R1.1 + R1.3 + Q1 records $A_{\text{fit}} \approx 2d$ at $d = 3, 4$ at the analytic level ([5, §4.1]). The formal-to-analytic sectorial-summability upgrade (Borel–Laplace radius identification at the SIARC stratum, [1, ch. 5]) is the jurisdiction of T1 Phase 3 (LANE-2 Item 2 sub-task 3-D authorisation [4, Item 2, sub-task 3-D]; gated on a separate relay).

[1] Term pending W21 canonical T1-Synth (LANE-1) rule5-vocabulary ratification per LANE-2 Item 4 verdict `DEFER_TO_W21` ([4, Item 4, L210–213]). The present note uses the term descriptively only.

- The borderline-locus posit (i′) as a structural mechanism on SIARC strata distinct from the $(1, b)$ corner remains an open future inquiry. The present note records the row-membership reading as a closure of the v1.0 §6 `q:mechanism` content for the V_{quad} $d = 2$ case under the extended enumeration; it does not rule out structural roles for (i′) elsewhere.
- An explicit PCF-1 v1.4 §6 row-table amendment that renders the $\deg a = 0$ row reading at the source-prose level is forward-pointed under the `PCF1-V13-RECONCILE` jurisdiction ([7, §6]). The PCF-1 v1.3 source `p12_pcf1_main.tex` (file SHA-256 E83BB377F297DBF0837FACBA257F227DF4579E6A3518C13
remains canonical and is unmodified by this note.

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AI Disclosure

During the preparation of this work the author used GitHub Copilot (Microsoft) and Anthropic Claude to assist with manuscript drafting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content.

References

[1] Ovidiu Costin. *Asymptotics and Borel Summability*, volume 141 of *Monographs and Surveys in Pure and Applied Mathematics*. Chapman and Hall/CRC, 2008. Cited in this follow-up note as a forward pointer to the formal-to-analytic sectorial-summability upgrade (T1 Phase 3 jurisdiction).

[2] Papanokechi. LANE-2 independent depth probe (p1–p5 findings). `siarc-relay-bridge`, `sessions/2026-05-06/T1-SYNTH-SUBSTITUTE-LANE2-051-Q1Q2Q4/independent_depth_probe.md`, 2026. File SHA-256 20764101FCDEA73A57EE92B80C97B3EAB87C579CEEBED611FF9D2E6087B3885D, 16698 bytes.

[3] Papanokechi. LANE-2 independent substrate verification (v1–v6 derivation log). `siarc-relay-bridge`, `sessions/2026-05-06/T1-SYNTH-SUBSTITUTE-LANE2-051-Q1Q2Q4/independent_substrate_verification.md`, 2026. File SHA-256 56063BF7BA8AD6A01A89FA30A3C61FCE68F4AAB0BA0DC7C8E561A81188D7B1F5, 15695 bytes.

[4] Papanokechi. LANE-2 six-item verdict (items 1–6 adjudication; `ACCEPT_WITH_REVISIONS` meta-verdict). `siarc-relay-bridge`, `sessions/2026-05-06/T1-SYNTH-SUBSTITUTE-LANE2-051-Q1Q2Q4/lane2_six_item_verdict.md`, 2026. File SHA-256 541663C69A5CE86B4F5D3799B04A0334C4A27E202DD9B3E2B80AFE16EE62B917, 18890 bytes.

[5] Papanokechi. A newton-polygon formal-level baseline for the Birkhoff–Trjitzinsky exponent on the SIARC polynomial continued-fraction wallis stratum. Zenodo. Concept DOI 10.5281/zenodo.20048196; v1.0 DOI 10.5281/zenodo.20048197, 2026. Source file SHA-256 6746692C517DC25238473E819527C5682465CDC9E1DEF69D1F6DF31C1014D51B, 38023 bytes; deposited at `siarc-relay-bridge/sessions/2026-05-06/T1-PHASE2-BASELINE-NOTE/bt_baseline_note.tex`. PDF SHA-256 23022F0DE77AC8388ED584B2196C0AB995CD8CF18B2DD71EFBC0488A0F6E5B7C, 409337 bytes.

[6] Papanokechi. PCF-1 v1.3: Polynomial continued fractions and the SIARC wallis stratum (16pp source). Zenodo. Concept DOI 10.5281/zenodo.19937196, 2026. Source file `p12_pcf1_main.tex`, SHA-256 E83BB377F297DBF0837FACBA257F227DF4579E6A3518C139E3146F17174BE301, 46349 bytes, 925 lines.

[7] Papanokechi. PCF-1 v1.3 § 6 V_{quad} row reframing – forward substrate (relay 066). `siarc-relay-bridge`, commit 9261c79, folder `sessions/2026-05-07/PCF1-V13-V_QUAD-ROW-REFRAMING-066/`, 2026. File SHA-256 79933B694DD2BF99793429B1A122A4BFF3260A42A93680886D5EF89B4E10FDCD, 24073 bytes.

[8] Papanokechi. PCF-2 cf_value implementation uniformity audit (13 impls) (relay 065). `siarc-relay-bridge`, commit 6a150b6, folder `sessions/2026-05-06/PCF2-CF_VALUE-AUDIT-9IMPLS-065/`, 2026. File SHA-256 16512BCC71C9A19ED59B808D8D070877F2160C5A7A062381C0FC9533182D3BF9, 11700 bytes.

[9] Papanokechi. PCF-2 v1.3: Higher-degree SIARC polynomial continued fractions (r1.1 + r1.3 + q1 harvest record). Zenodo. Concept DOI 10.5281/zenodo.19963298, 2026.

[10] Papanokechi. Phase A WZ-balance table – supplementary $\deg a = 0$ row (relay 064). `siarc-relay-bridge`, commit 6a150b6, folder `sessions/2026-05-06/PHASE-A-DEG_A-ZERO-SUPPLEMENTARY-064/`, 2026. File SHA-256 80E28568FF142B1A81C7C443368349A07A8235CE122D6C6140F88AA0D0706097, 16792 bytes.

[11] Papanokechi. T1-SYNTH-SUBSTITUTE-LANE2-051-Q1Q2Q4: Canonical LANE-2 re-arbitration (`ACCEPT_WITH_REVISIONS` + 6-item adjudication). `siarc-relay-bridge`, commit `dee3c01`, folder `sessions/2026-05-06/T1-SYNTH-SUBSTITUTE-LANE2-051-Q1Q2Q4/`, 2026.

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